

Louis Bersine

OT Infrastructure & Controls Architect

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EXECUTIVE SUMMARY

OT Infrastructure & Controls Engineer specializing in brownfield network stabilization, industrial cybersecurity alignment, and high-availability manufacturing systems. Proven record of mitigating six-figure downtime exposure and eliminating contractor dependency through internal architecture leadership.

CORE STRENGTHS

- Brownfield OT network stabilization (VLANs, IGMP, EtherNet/IP, CIP Sync / IEEE-1588 PTP)
- Rockwell controls + visualization (ControlLogix/GuardLogix, FactoryTalk SE/ME, AssetCentre, OPC DA/UA)
- OT governance, disaster recovery, and cyber-aligned engineering environments
- Virtualized toolchain standardization and reproducible engineering workstation baselines
- Automation tooling and scripting to improve reliability, visibility, and engineering throughput

PROFESSIONAL EXPERIENCE

Kellanova (Kellogg)

Controls Engineer / OT Network Manager

Grand Rapids, MI | Dec 2023 - Present

- Resolved CIP Sync/PTP grandmaster conflict impacting motion layer; restored deterministic timing plant-wide in ~6 hours, mitigating ~\$3,500/hr downtime exposure.
- Implemented PLC change tracking, automated backups, and OPC restoration across undocumented servers; avoided ~\$40,000–\$80,000 in external recovery costs.
- Delivered GuardLink safety upgrade correcting single-channel e-stop risk using existing infrastructure; ~\$30,000 vs. \$200,000 proposal (~\$170,000 capital avoided).
- Standardized virtualized Rockwell toolchains and reproducible engineering image baselines; improved supportability/cyber alignment with ~\$1,400 savings per workstation.
- Partnered with IT on OT cybersecurity/governance: segmented access, AssetCentre automated backups, and domain-integrated engineering environments.
- Stabilized legacy industrial Ethernet (VLANs, IGMP, CIP/IEEE-1588 PTP); reduced master clock instability and downtime risk.
- OT escalation lead for network/automation incidents; coached teams on EtherNet/IP and OPC UA.

Proficient Machine & Automation

Controls Engineer

Byron Center, MI | Sep 2019 - Sep 2023

- Led electrical standards migration (SolidWorks → AutoCAD Electrical): standardized symbols, PLC modules, title blocks, and wire-numbering macros; protected ~\$3M–\$8M annual automation revenue.
- Built VBA-based HMI platform replacing licensed third-party visualization; avoided ~\$15,000–\$40,000 per deployment.
- Created Excel-driven PLC tag auto-generation for badge systems; reduced commissioning labor and protected ~\$20,000–\$50,000 across deployments.
- Designed hardened Raspberry Pi remote access using non-writable architectures, certificate-based auth, and encrypted client connectivity.
- Engineered PLC/HMI/robotic automation software for deterministic motion control and throughput stability.

Roman Manufacturing

Controls Engineer

Wyoming, MI | Aug 2018 - Sep 2019

- Developed VBA tools leveraging Autodesk Inventor models to automate cable measurement, improving bid accuracy and material planning.
- Recovered FANUC CNC system without OEM documentation by reverse-engineering control behavior; eliminated contractor dependency and restored production stability.
- Managed estimated \$50,000–\$300,000+ per project in parts/services/equipment spend while modernizing systems to meet machinery safety standards.
- - Worked directly with clients to manage requirements for building and testing welding equipment designs (~\$100,000–\$400,000 project value).
- Modernized systems to meet machinery safety standards and collect metrics; managed estimated \$50,000–\$300,000+ per project in parts/services/equipment spend.
- Engineered PLC-based automation systems in high-heat refinery environments with reliable temperature acquisition and custom signal integration.

The Party Zone VR LLC

Founder / System & Security Architect

Michigan (Remote) | Sep 2022 - Present

- Architected custom software systems for immersive online music experiences; contracted by Universal Music Group Sweden and Denison University.
- Designed scalable infrastructure supporting interactive digital performances and creator-driven engagement.
- Built and managed virtual operations frameworks for live online concerts, including community coordination and high-engagement Discord environments.
- Developed advanced VRChat products and interactive assets to expand creator monetization and enhance social VR experiences.
- Built low-latency live performance telemetry + multi-user synchronization in VR using OSC + TCP messaging; reduced manual operator load, improved show reliability, and enabled repeatable live performance workflows.
- Implemented an unattended, self-healing interactive “show” orchestration system that restarts safely and generates fresh content via AI prompts + AI voice playback; reduced operational babysitting.
- Built a full web + API prototype for ad submission and VR display integrations (Metaplaza VR), validating end-to-end delivery flow (concept later sunset).

EDUCATION

Michigan Technological University — BSEE, Electrical Engineering | Class of 2015

TECHNICAL SKILLS

Controls Engineering:

AutoCAD, Studio 5000, RSLogix 500, FactoryTalk View ME/SE, FactoryTalk AssetCentre, Fanuc/Denso/IAI Robots, PowerFlex VFD, Kinetix, Cognex/Keyence Vision, Atlas Copco, Siemens S7, Beckhoff TwinCAT

OT Networking & Administration:

TCP/IP, PTP, NTP, EtherNet/IP, CANbus, VPNs, Firewalls, Active Directory, Claroty, OSIsoft PI, Kepware, Ignition, Wireshark, Cisco CLI

Programming & Automation:

PowerShell, Python, AutoLISP, C#, C++, JavaScript, Power Automate, VBA, MATLAB, XML, JSON